ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** two terminals	Positive and negative.
2	**(b)** open circuit	No current.
3	**(d)** switch off the supply before fixing a plug	Safe.
4	**(a)** positive terminal	Positive.
5	**(c)** closed circuit	Current flows.
6	**(c)** hot	Heating effect.
7	**(c)** copper	Metal.
8	**(b)** chemical energy into electrical energy	Cell energy.
9	**(a)** an electromagnet	To lift iron.
10	**(a)** rubber	Not a conductor.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cell and wires (1); Switch and bulb (1).	1+1
12	A safety device with a thin wire (1); Melts when current is excessive (1).	1+1
13	Direct contact of live and neutral wires (1); Large current flows (1).	1+1
14	Copper (1); Aluminium (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell connected to bulb by wires (1); Switch in series (1); Closed switch makes bulb glow (1).	1+1+1
16	Don't touch bare wire (1); Don't use wet hands (1); Switch off before repair (1).	1+1+1
17	High resistance (1); More heat (1); High melting point (1).	1+1+1
	OR: Conductors: copper, iron, aluminium (1); Insulators: rubber, plastic, glass (1); Conductors let current pass (1).	1+1+1
18	Short circuit (1); Overloading (1); Poor wires (1).	1+1+1
	OR: Fuse melts, MCB trips (1); Fuse must be replaced (1); MCB can be reset (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cell (1); Wires (1); Switch (1); Bulb (1); Closed circuit allows current (1).	1+1+1+1+1
	OR: Iron core (1); Coil (1); Armature and hammer (1); Gong (1); Repeating action (1).	1+1+1+1+1
20	Make a circuit with a gap (1); Place the material in the gap (1); Bulb glows: conductor (1); Bulb doesn't glow: insulator (1); Examples (1).	1+1+1+1+1
	OR: Use dry hands (1); Don't touch bare wires (1); Switch off before repair (1); Use proper plugs (1); Don't overload (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.
- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** electricity and magnetism	Magnetic effect.
2	**(b)** insulated wire and a cell	Coil needed.
3	**(d)** start or stop current	On/off.
4	**(d)** small magnet	Points north-south.
5	**(c)** an electromagnet	Pulls hammer.
6	**(b)** amperes	A.
7	**(b)** nichrome	High resistance.
8	**(c)** several cells	Cells together.
9	**(c)** current flows	Current makes it magnetic.
10	**(c)** a magnetic field	Magnetic.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Opens or closes the circuit (2).	2
12	Magnetic effect (2).	2
13	Rubber (1); Plastic (1).	1+1
14	Ampere (1); Symbol A (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell: long and short lines (1); Switch: gap with a line (1); Bulb: circle with cross (1).	1+1+1
16	More turns (1); More current (1); Use a soft iron core (1).	1+1+1
17	Heating effect (1); Magnetic effect (1); Chemical effect (1).	1+1+1
	OR: Bell (1); Crane (1); Scrap separation (1).	1+1+1
18	It deflects (1); Shows magnetic effect (1); Direction changes if current reversed (1).	1+1+1
	OR: Has two terminals (1); Converts chemical energy to electrical (1); Several cells form a battery (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take an iron nail (1); Wind insulated wire (1); Connect to a cell (1); Test with pins (1); Disconnect to demagnetise (1).	1+1+1+1+1
	OR: Heater: heating (1); Fan: motion (1); Bulb: light (1); Bell: magnet (1); Cell: chemical (1).	1+1+1+1+1
20	Wire, cell, switch, compass (1); Close the switch (1); Needle deflects (1); Magnetic field around wire (1); Reverse current reverses deflection (1).	1+1+1+1+1
	OR: Electromagnet needs current (1); Strength can be changed (1); Poles can be reversed (1); Bar magnet is permanent (1); Electromagnet is switched off (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.
- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** salt solution	Contains ions.
2	**(c)** melts when the current is too high	Protective.
3	**(b)** avoid shocks	Insulation.
4	**(d)** too many appliances on one socket	Excess current.
5	**(a)** electric iron	Heats.
6	**(a)** heating of its filament	Heat and light.
7	**(a)** fire	Big current.
8	**(b)** tungsten	High melting point.
9	**(a)** miniature circuit breaker	Safety device.
10	**(c)** switched off	No current.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heating effect (2).	2
12	Coil on an iron core (1); Magnet when current flows (1).	1+1
13	Tungsten (1); Nichrome (1).	1+1
14	Plastic is an insulator (1); To avoid shocks (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Iron (1); Heater (1); Toaster (1).	1+1+1
16	So that all current passes through it (1); It melts on high current (1); Protects the circuit (1).	1+1+1
17	It loses magnetism (1); Iron core falls demagnetised (1); Used in cranes to release load (1).	1+1+1
	OR: Current flows through the filament (1); Filament heats up (1); It gives light (1).	1+1+1
18	Open: gap, no current (1); Closed: complete path, current flows (1); Switch controls it (1).	1+1+1
	OR: Water conducts (1); Current can pass through the body (1); May cause a shock (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Current heats a conductor (1); More resistance, more heat (1); Heater (1); Iron (1); Fuse (1).	1+1+1+1+1
	OR: Short circuit: live meets neutral (1); Overloading: many appliances (1); Both give high current (1); Causes fire (1); Fuse or MCB prevents (1).	1+1+1+1+1
20	Thin wire of low melting point (1); In series (1); Melts on high current (1); Protects appliances (1); Replace after it burns (1).	1+1+1+1+1
	OR: Filament: tungsten (1); Coil: nichrome (1); Filament gives light (1); Coil gives heat (1); Both heat up by current (1).	1+1+1+1+1

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- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.
- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

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